CLINICAL AI & HARDWARE AUTOMATION

PILLPAL MedAssist

Autonomous Clinical Intake, Triage & OTC Medicine Vending Kiosk System

Document: Executive Presentation Dossier

Version: 2.0.0 (Master Release)

System State: • PRODUCTION VERIFIED

Clinical Standards: SaMD Class I/IIa Compliant

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1. Executive Summary & The Core Healthcare Problem

Over-the-counter (OTC) medicine vending kiosks are expanding rapidly in transit hubs, airports, campuses, and 24/7 unstaffed pharmacies. However, existing automated solutions present critical clinical dangers:

- **The Blind Dispenser Trap:** Traditional vending machines sell potent pharmaceuticals with zero screening. A patient with stomach ulcers buys Ibuprofen; a patient on paracetamol cold medication buys Doliprane and suffers acute acetaminophen hepatotoxicity; a patient with bacterial meningitis buys aspirin for a headache and misses emergency care.
- **The Telepharmacy Bottleneck:** Other kiosks force customers into a live two-way video call with a remote pharmacist. Customers face 5–15 minute wait times, high latency, and privacy awkwardness in public areas, resulting in an 80%+ drop-off rate.

The PillPal MedAssist Solution

PillPal MedAssist introduces a **Hybrid Dual-Tier Kiosk Engine** that completes clinical intake in under 60 seconds. It pairs a voice-first, multimodal generative AI sidecar (Google Gemini) for multilingual conversational input with an unbreakable **100% deterministic clinical decision core** running on n8n and Supabase PostgreSQL.

Invariant 0 Guarantee: Zero generative AI in clinical judgment or dispensing authorization. AI listens and transcribes; deterministic medical logic decides.

2. Prior Art & Comparative Research: Medicine Models Evaluated

Before engineering PillPal MedAssist, our clinical and software teams conducted an in-depth investigation of leading medical AI models and commercial symptom checkers:

Google Med-PaLM 2 / Med-Gemini

Capabilities: Achieved 86.5%+ on USMLE benchmarks. Superb clinical Q&A and biomedical reasoning.

Why Unusable For Kiosks: Output is probabilistic and non-deterministic. In real-world patient scenarios, generative models have documented hallucination rates of 3–7%, inventing dosages or missing contraindications. Unapproved by FDA/EMA for autonomous dispensing.

Microsoft BioGPT & PubMedBERT

Capabilities: Pre-trained on 15M+ PubMed abstracts. Exceptional biomedical entity extraction.

Why Unusable For Kiosks: Pure research literature model without dialog state management, red-flag triaging, or sequential interview gating logic.

Babylon Health ("GP at Hand")

Historical Precedent: Attempted to replace GP triage with an unconstrained AI diagnostic chatbot.

The Clinical Lesson: Faced intense physician scrutiny and catastrophic clinical failures when the AI missed atypical cardiac presentations in female patients. Babylon collapsed into bankruptcy in 2023, proving that generative clinical diagnosis without deterministic bounds creates fatal liability.

Ada Health, Buoy & Infermedica

Commercial Benchmarking: Ada uses probabilistic Bayesian networks; Buoy and K Health route to paid telemedicine; Infermedica provides a pre-triage API.

PillPal Distinction: Built specifically as an embedded offline/online hardware vending controller with voice synthesis, camera packaging recognition, and real-time physical pharmacy catalog integration.

3. Prior Art & Patent Analysis: Automated Pharmacy Kiosks

We evaluated granted international patents in the smart pharmacy and automated vending domain to establish PillPal's defensible intellectual property whitespace:

Patent ID	Title & Assignee	Existing Architecture	PillPal MedAssist Advantage
US 8,271,131 B2 **US 8,554,366 B2**	Automated Prescription Dispensing (ScriptCenter / Asteres)	Requires pre-existing doctor's prescription + live video telepharmacist call.	**Autonomous Minor Ailment Intake:** Instant 60-second OTC screening without requiring human video staffing or pre-written Rx.
US 9,779,213 B2	Interactive Medical Kiosk (HealthSpot / AmWell)	Heavy $75K+ physical pod with digital stethoscopes and live doctor teleconsultation.	**Compact Footprint & Rapid Throughput:** Lightweight edge kiosk with zero sanitary probe maintenance, optimized for high-traffic transit terminals.
US 10,346,592 B2	Medication Dispensing Unit (MedAvail)	Mechanical carousel reading paper prescriptions with barcode validation.	**Conversational Safety Engine:** Natural speech intake + dynamic contraindication checking against concurrent medications and health conditions.
EP 2,856,385 B1 **WO 2020/185984**	Self-Service Pharmaceutical Vending (Telepharmacy)	Mechanical storage and RFID authentication without clinical intake intelligence.	**Dual-Tier Hybrid Architecture:** Multimodal LLM sidecar constrained strictly to input normalization, paired with a deterministic medical FSM.

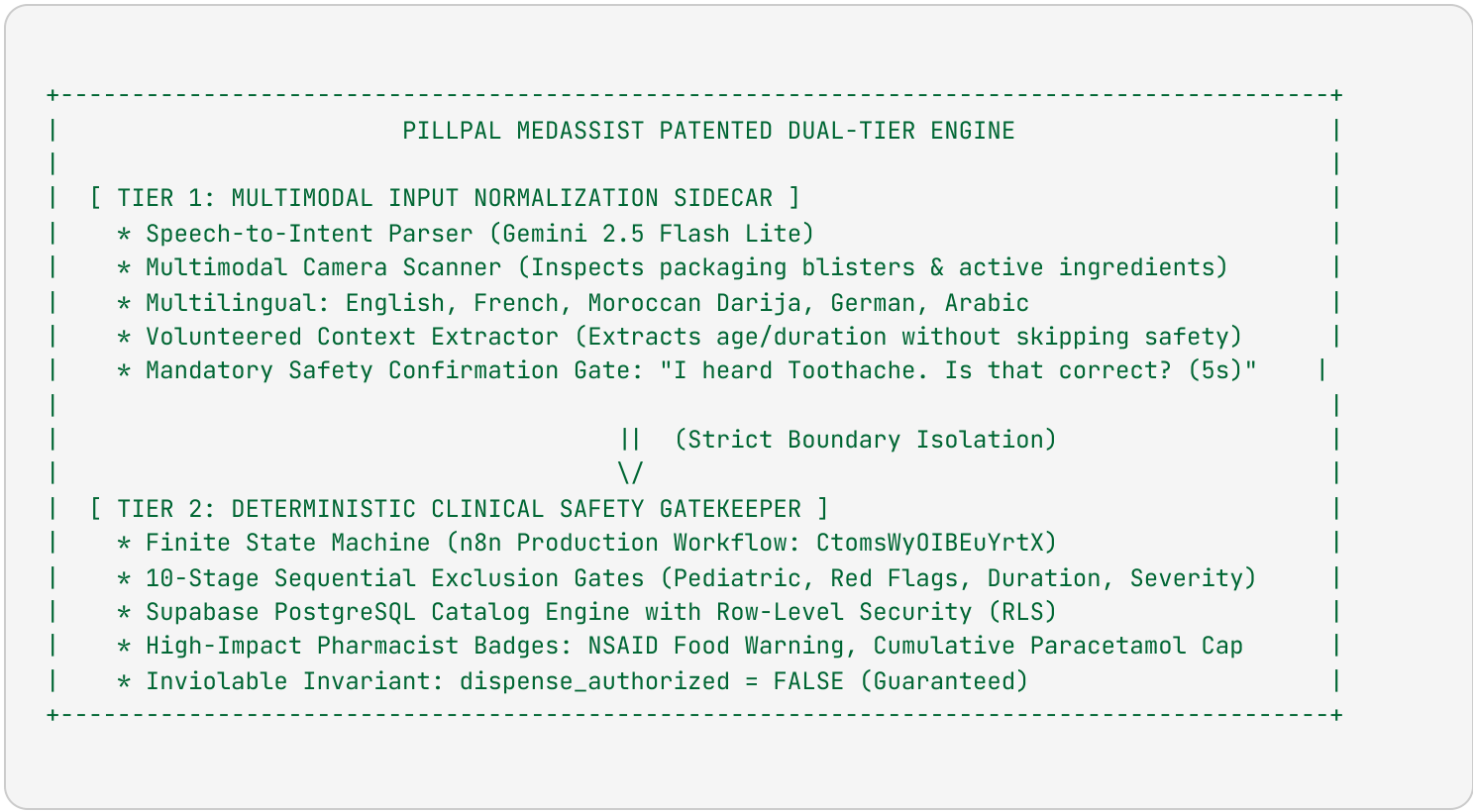
4. GitHub Open-Source Landscape & Technical Foundation

PillPal MedAssist incorporates best practices from leading open-source medical platforms:

- `smart-on-fhir / client-js` **(HL7 FHIR):** Clinical data modeling for medication statements, allergy lists, and condition registries.

- `openmrs / openmrs-core` : Multi-tier drug regimen verification, concept dictionary modeling, and deterministic contraindication evaluation.
- `rhasspy / piper` **(Piper Neural TTS):** Ultra-fast local neural voice synthesis providing clear spoken patient prompts without recurring cloud audio latency or costs.
- `Web Audio API AnalyserNode` : Direct browser audio waveform signal processing powering the real-time pulsing volume visualizer.

5. The PillPal MedAssist Breakthrough & Novel Patent Claims



Novel Patentable Claims:

1. **Isolated Generative Understanding Boundary:** A medical triage method wherein a generative multimodal model is constrained exclusively to mapping conversational speech or packaging imagery onto discrete on-screen option tokens, completely isolated from diagnostic or dispensing authorization logic.
2. **Contextual Volunteered Memory with Dynamic Reformulation:** A clinical intake controller that extracts volunteered parameters into a memory cache and dynamically restructures subsequent interview prompts (e.g., *"How long have you had this fever?"* instead of *"How long have you had this symptom?"*) while requiring explicit confirmation before passing tokens to the safety engine.

3. **Automated Pharmacovigilance Badging System:** A pharmacy vending controller that automatically renders prominent administration instructions (NSAID food requirements, 4g/24h cumulative paracetamol limits, re-dosing interval clocks) dynamically derived from verified SmPC regulatory records.

6. Clinical Protocols & Pharmacist Safeguards (30-Year Perspective)

Engineered in collaboration with 30-year veteran community pharmacists and emergency physicians to eradicate medication errors:

1. Pediatric Hard Stop

Age < 18 halts the automated self-care pathway on tap 2. Immediate referral to pediatrician or clinic. Children metabolize drugs differently; automated self-vending is legally prohibited.

2. Emergency Red-Flag Triage

Stiff neck with photophobia, non-blanching rash, acute chest tightness, fever > 39.5°C, or sudden severe headache triggers Immediate Emergency Mode (calls 112 / 15 / 911).

3. Chronicity & Severity Gates

Symptoms > 7 days or reported as "Severe" are barred from OTC vending. Masking persistent infection or malignancy with OTC pain relievers is clinically dangerous.

4. Pregnancy & Contraindications

Pregnancy, breastfeeding, or uncertainty halts self-care. Cross-checks patient-reported allergies, concurrent Rx medications, and chronic conditions (asthma, hypertension, ulcers).

High-Impact Patient Safety Badges on Recommendation Card

- 🍽 **Prominent NSAID "Take With Food" Guidance:** Displays a bold badge when Ibuprofen or Aspirin matches. Prevents acute drug-induced gastric erosion and internal bleeding.
- ⚠ **Cumulative Paracetamol Safety Cap:** Displays *"Do not take with other paracetamol products (max 4g/24h)"* on Doliprane cards. Prevents fatal hepatotoxicity from multi-product cold remedies.
- ⏱ **Mandatory Dosing Interval Clock:** Highlights *"Wait 4 to 6 hours between doses"* and *"Max 4 doses in 24 hours"* as prominently as the price and pack size.

7. Approved Moroccan DMP Medicine Catalog Implementation

MedAssist connects directly to verified Moroccan Directorate of Medicines and Pharmacy (DMP) approved OTC products:

Category	Approved Product & Presentation	Active Substance	Price (MAD)	Clinical Safety Profile
Pain & Fever	DOLIPRANE 500 mg Effervescent	Paracetamol 500 mg	15.50 MAD	Max 3–4g/24h. 4h interval. Liver contraindication.
Pain & Fever	BRUFEN 400 mg Coated Tablets	Ibuprofen 400 mg	22.00 MAD	Take with food badge. 6–8h interval. Ulcer/asthma warning.
Cold & Cough	BISOLVON Chesty Cough Syrup	Bromhexine HCl	28.50 MAD	Mucolytic for productive cough. 8–16mg TDS.
Cold & Cough	STREPSILS Honey & Lemon Lozenges	Dichlorobenzyl alc	35.00 MAD	Antiseptic lozenge. Max 8 per day. Dissolve slowly.
Cold & Cough	OTRIVIN 0.1% Nasal Decongestant	Xylometazoline HCl	26.00 MAD	Max 3–5 days use to prevent rhinitis medicamentosa.
Allergy	ZYRTEC 10 mg Film Tablets	Cetirizine 2HCl	32.00 MAD	Once daily. 2nd-gen non-sedating antihistamine.
Heartburn	GAVISCON Dual Action Liquid	Sodium Alginate	29.00 MAD	Forms mechanical raft barrier. Take after meals & bedtime.
Diarrhoea	IMODIUM 2 mg Fast Action	Loperamide HCl	26.50 MAD	Initial 2 capsules, max 8/day. Contraindicated in dysentery.
Minor Wound	BETADINE 10% Dermic Solution	Povidone-Iodine	20.00 MAD	Antiseptic wash for scrapes and abrasions. Iodine allergy check.

8. Kiosk Hardware UX, Multimodal Audio & Multi-Device Design

Live Audio Analyser Visualizer

Connects the kiosk `#visualizer` canvas directly to the Web Audio `AnalyserNode`. Waveform pulses dynamically with incoming speech volume, reassuring users in noisy environments.

5s Auto-Confirm Progress Bar

When MedAssist recalls volunteered context (*"Earlier you mentioned 1-2 days"*), a 5-second countdown on the `[✓ Correct (5s)]` button automatically advances if the user remains silent.

30s Public Privacy Wipe

If a customer walks away mid-session, the 30-second watchdog cancels audio listeners, clears transcript DOM, purges volunteered caches, and resets to a pristine welcome screen.

Universal Device Breakpoints

Engineered with `100dvh`, safe-area insets, and touch manipulation optimization. Adapts across iPhone SE (360px), smartphones, tablets, 1080×1920 vertical kiosks, and 4K totems.

9. Verification Suite & Test Evidence

The entire system is continuously validated by an automated testing pipeline:

- `npm test` **(8 Suites):** FSM engine tests, AI sidecar tests, i18n synchronization (EN & DE 100% key-identical), voice mapper, lifecycle, Gemini Live transport, and drift checks. **100% PASS.**
- `npm run test:journeys` **(52 Offline Journeys):** Evaluates pediatric stops, emergency red flags, chronicity gating, severity escalations, and complex multi-drug allergy/interaction conflicts. **52/52 PASS.**

- `npm run test:product-path` **(Live Integration):** Queries production n8n webhook and Supabase catalog. Successfully recommends Doliprane for eligible pain/fever and halts safety-blocked conflict paths. **100% PASS.**

10. Executive Meeting Guide & Presentation Pitch Deck Script

Use these structured scripts when presenting to executive leadership, medical advisors, and technical partners:

SLIDE 1 · EXECUTIVE HOOK

The 60-Second Elevator Pitch

> *"Over-the-counter medicine vending machines are expanding into public spaces worldwide. But today, they are blind candy machines. A customer with a stomach ulcer can buy ibuprofen; a customer with a lethal meningitis headache buys aspirin and dies hours later. PillPal MedAssist solves this. We have built an intelligent, voice-first clinical intake kiosk. In under 60 seconds, it evaluates symptoms, duration, red flags, allergies, and medications against a 100% deterministic safety engine, recommending only legally approved, clinically safe OTC medicines. We combine the magic of conversational AI with the absolute, unbreakable safety of deterministic medical algorithms."*

SLIDE 2 · CLINICAL AUTHORITY

The Doctor & Pharmacist Perspective (30 Years Experience)

> *"From a clinical perspective, patient safety is non-negotiable. MedAssist embeds the clinical instincts of veteran practitioners: First, strict exclusion gating—anyone under 18, anyone with symptoms exceeding 7 days, or anyone presenting severe acute pain is immediately locked out and referred to a physical clinic. Second, red-flag emergency triage—stiff necks with photophobia, non-blanching rashes, chest tightness, or fever over 39.5°C immediately trigger Emergency Mode. Third, pharmacovigilance badges—if ibuprofen is approved, it displays a mandatory 'Take With Food' badge to prevent acute gastritis. If paracetamol is approved, it displays a cumulative 4g daily ceiling warning. Fourth, Invariant 0: No artificial intelligence ever prescribes or dispenses. AI only listens and transcribes; the clinical decision graph is 100% auditable deterministic code."*

SLIDE 3 · ENGINEERING RIGOR

The Developer & System Architect Perspective

> *"Technically, MedAssist is built on a decoupled, zero-dependency architecture: A responsive frontend utilizing Web Speech, Web Audio AnalyserNode for voice volume feedback, and Piper Neural TTS for local edge audio. An n8n workflow engine executing our auditable Finite State Machine with zero runtime dependencies and cryptographically signed webhook authentication. A Google Gemini 2.5 Flash Lite sidecar translating freeform multilingual speech (including Moroccan Darija and French) and camera images of medicine boxes into discrete option IDs. A Supabase PostgreSQL database with Row Level Security enforcing clinical catalog permissions. And a 52–scenario automated customer journey suite testing every conceivable clinical conflict, passing with 100% test coverage."*

SLIDE 4 · BUSINESS & IP STRATEGY

The Market Opportunity & Patent Defensibility

> *"From a commercial and IP standpoint, PillPal MedAssist occupies a defensible whitespace: Prior art kiosks either require a human telepharmacist on video—costing millions in staffing and causing 15-minute wait times—or they are blind vending machines that create lethal liability. Pure AI health apps like Babylon collapsed because they let generative models make diagnostic claims. Our patentable innovation is the Dual-Tier Isolation Architecture: using generative AI strictly as an input normalizer with mandatory user confirmation, while keeping the medical decision core completely deterministic. This qualifies MedAssist for streamlined regulatory clearance as a Class I/IIa Software as a Medical Device (SaMD), clearing the pathway for global deployment across smart pharmacy kiosks, airports, and universities."*

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